## CBC-MAC Forgery Attack

## 30. Write a C program for CBC MAC of a oneblock message X, say T = MAC(K, X), the adversary immediately knows the CBC MAC for the two-block message X || (X ⊕ T) since this is once again.

## Aim:

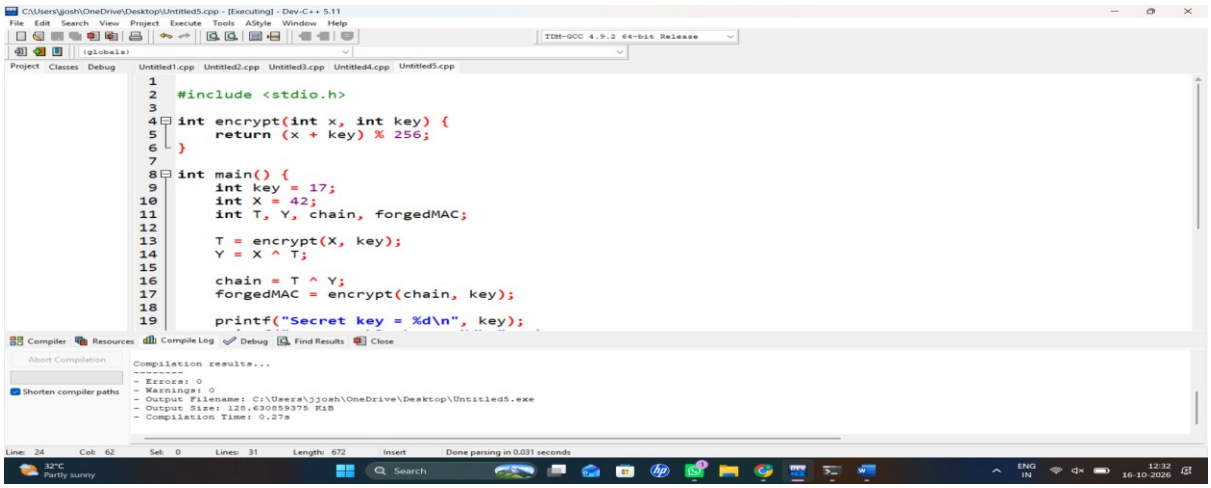
To write a C program demonstrating how an attacker can construct a two-block message with the same CBC-MAC as a known one-block message.

## Algorithm:

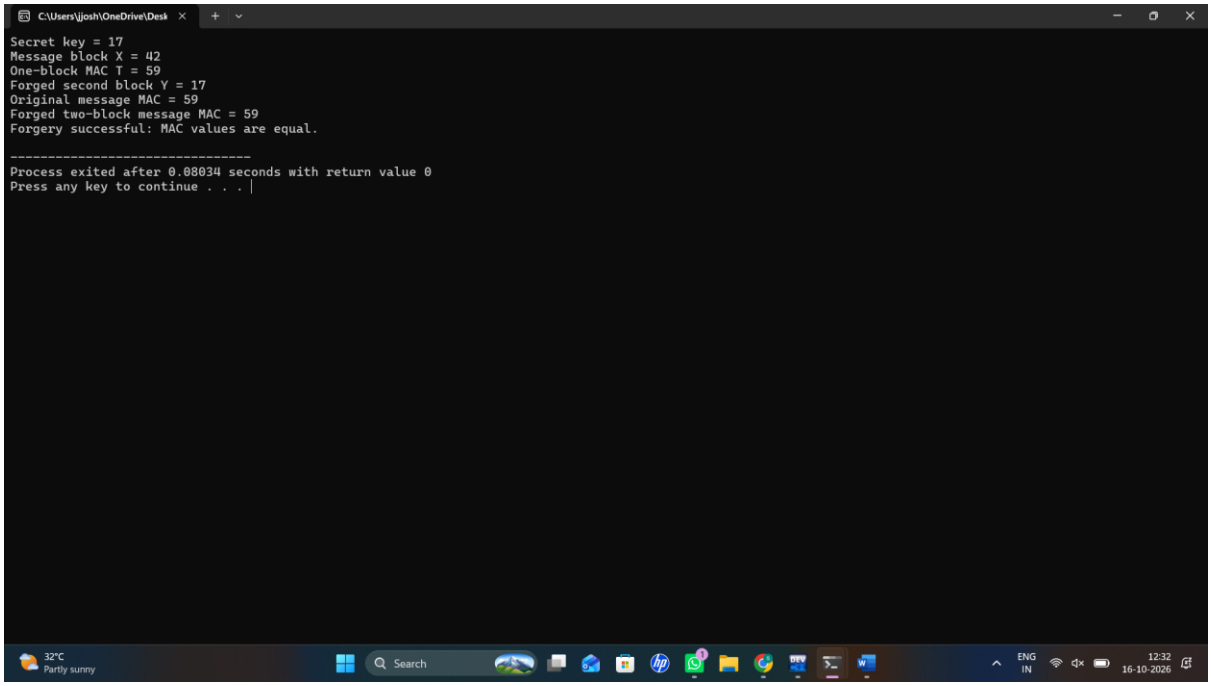
1. Choose a message block \(X\) and a secret key \(K\).
2. Calculate the one-block CBC-MAC: \(T=E_K(X)\).
3. Construct a second block \(Y=X\oplus T\).
4. Form the two-block message \(X\parallel Y\).
5. Calculate its CBC-MAC using the CBC chaining operation.
6. Verify that the two-block message produces the same MAC \(T\).

## Program:

The following program uses a simple toy encryption function for demonstration, not a real block cipher.



## Output:




## Result:

The program demonstrates the CBC-MAC forgery property. Given the MAC \(T\) for a one-block message \(X\), an attacker can construct \(X\parallel(X\oplus T)\), which produces the same MAC under the CBC-MAC construction. This demonstrates why basic CBC-MAC should not be used directly for variable-length messages without an appropriate secure construction.